

AI-DRIVEN CLIMATE-AWARE DECISION SUPPORT AND RISK FORECASTING SYSTEMS FOR SRI LANKA

PROJECT ID : R26-IT-046





Every Year Floods

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Comments	Our Improvements
Minor issues and inconsistencies were observed in the model.	Fine-tuned the model parameters and evaluated different training approaches to reduce prediction errors and improve overall accuracy.

The Problem and It's Impact

Problem

- People **do not receive** early flood warnings.
- People are **not aware** of dengue risk levels in their surrounding areas.
- Farmers **do not receive** specific advice related to weather and flood conditions.
- People find it difficult to identify **safe travel routes** during floods.



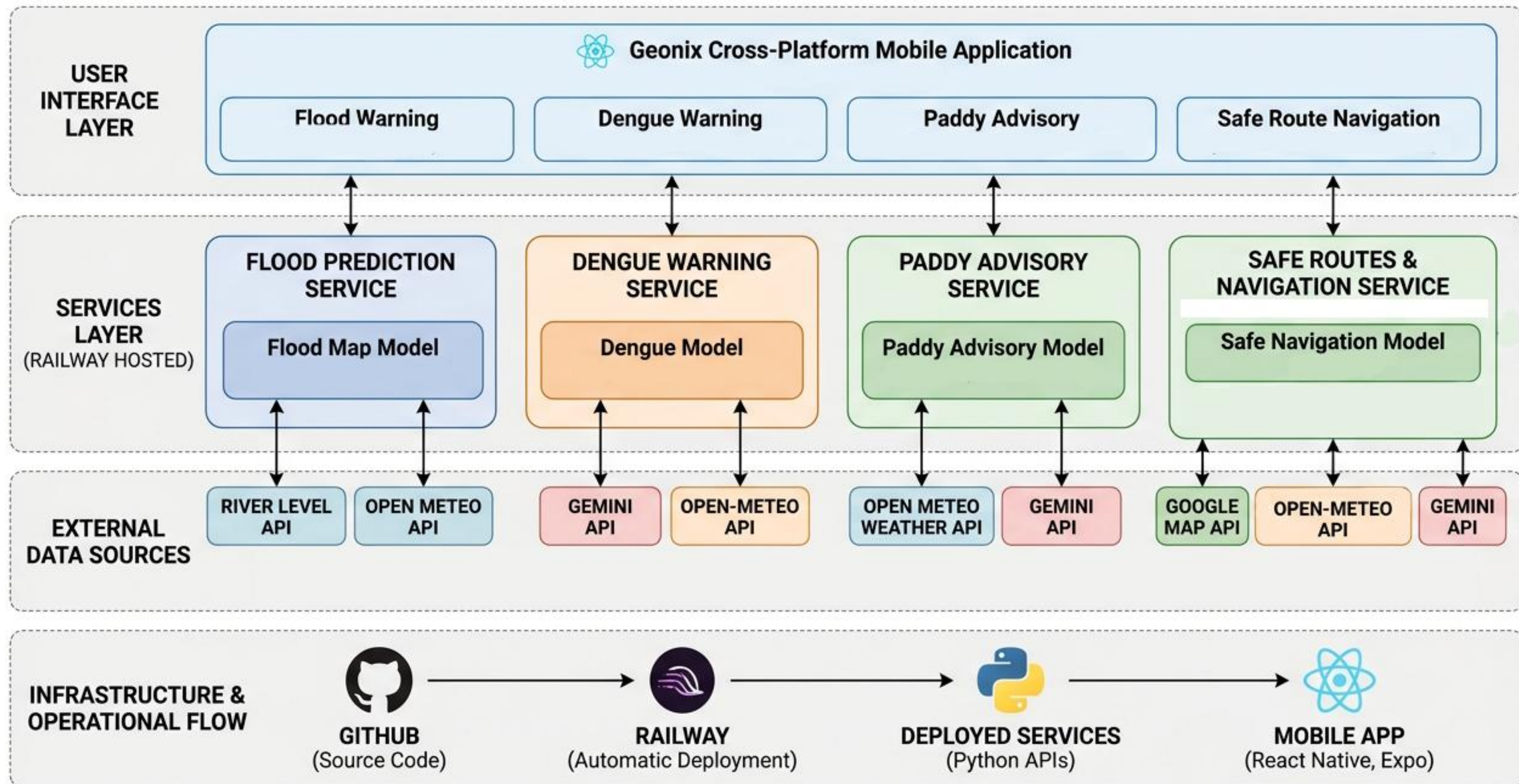
Impact

- Improves public safety
- Increases awareness of dengue-prone areas
- Supports farmers with location-based advice
- Helps people identify safer travel routes

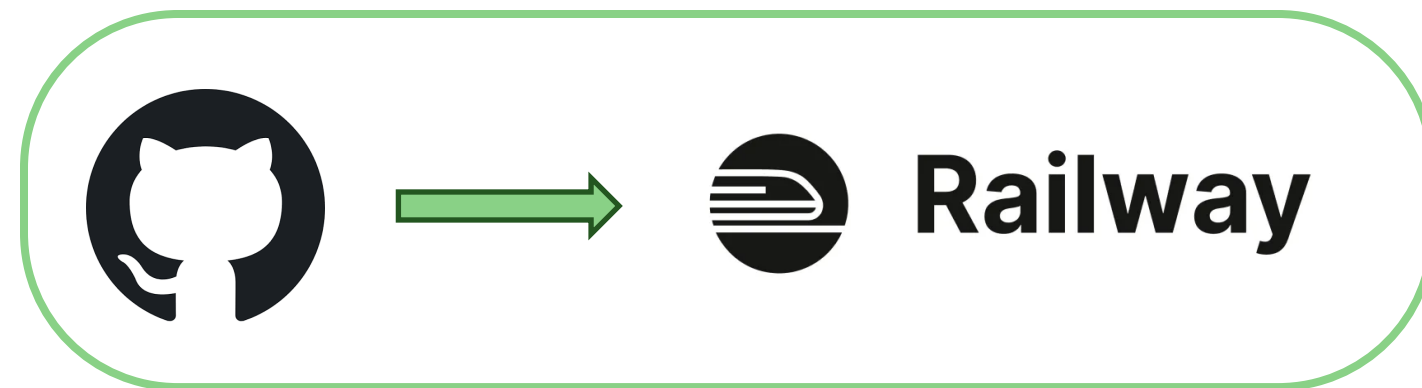
Design Excellence

- Easy-to-use interface for flood information access
- Clear interactive flood risk map
- Accurate predictions using trained ML models
- Real-time alerts with updated data

SYSTEM DIAGRAM



INTEGRATION AND COMPLETION



- **GitHub Collaboration** - Centralized repository for team collaboration, version control, and code integration.
- **Backend Deployment** - Fast API backend deployed on Railway for accessible API services.
- **End-to-End Integration** - Connected the ML model, backend API, and frontend to deliver the prediction workflow.

Non-Functional Requirement

- **Performance & Availability:** Since the backend has been deployed, the system's response time and availability have improved.
- **Usability:** The mobile application should provide a simple and user-friendly interface.
- **Responsiveness:** The mobile application's interface should respond quickly to user interactions.

Commercialization

API-as-a-service



Sell metered or subscription access to flood, dengue, paddy, and safe-navigation endpoints to other developers, NGOs, or agri-tech startups who don't want to build their own models.

Institutional
licensing

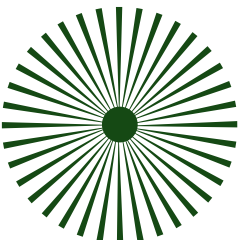
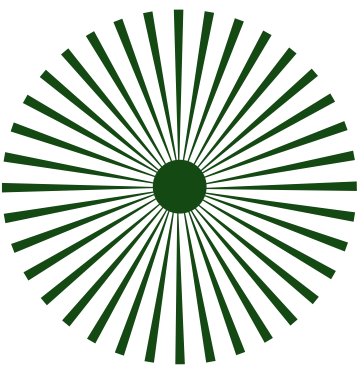


License the flood, dengue, and paddy modules directly to government bodies (Irrigation Dept, Ministry of Health, Dept of Agriculture) or NGOs that already have mandates around these exact hazards.

Freemium
consumer app



Keep the current mobile app free for individuals (farmers, commuters, residents) and monetize through premium features or ads, growing usage data that makes the underlying models more valuable over time.



THANK YOU ...

